

PROJECT REPORT

Title

EV GUARD – IoT Based Smart EV Charging Monitoring and Protection System

Abstract

The increasing adoption of Electric Vehicles (EVs) requires intelligent charging systems that ensure safe, efficient, and reliable charging. This project, **EV GUARD**, is an IoT-based smart EV charging monitoring and protection system developed using ESP32 and Blynk IoT. The system continuously monitors charging conditions such as voltage, simulated current, and temperature while displaying real-time information on a cloud dashboard. It also provides visual and audible alerts during different charging stages and automatically stops charging under abnormal conditions, making the charging process safer and more efficient.

1. Introduction

Electric Vehicles are becoming one of the major solutions for sustainable transportation. However, safe battery charging is an important challenge. Overheating, abnormal voltage, and charging faults can reduce battery life and create safety risks.

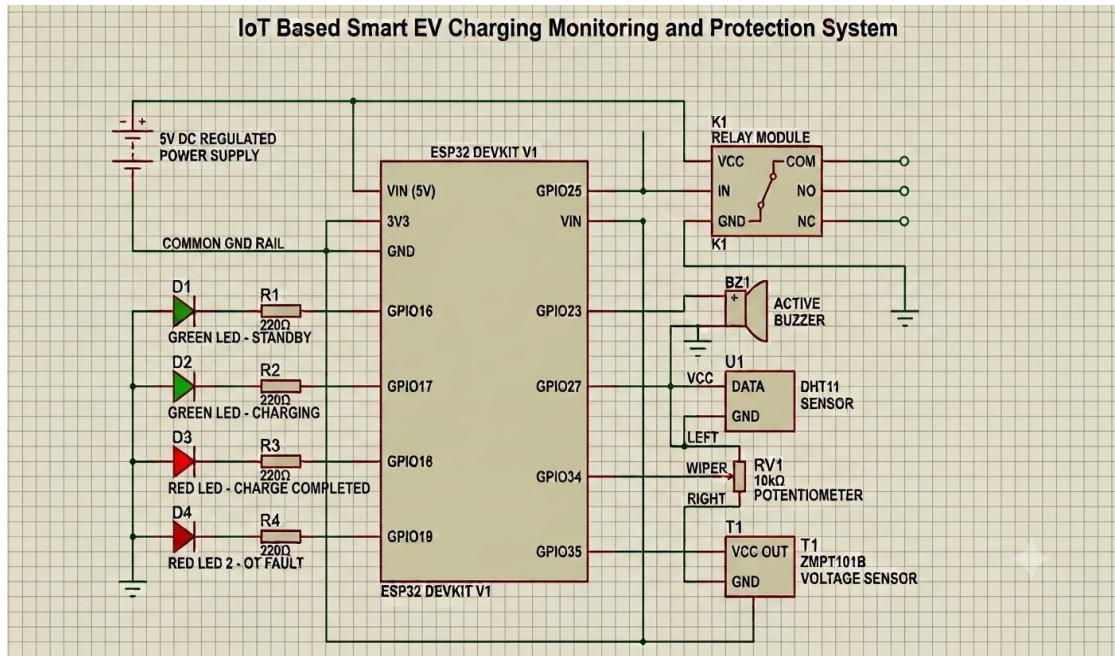
The EV GUARD project addresses these issues by providing real-time monitoring, remote visualization, and automatic protection using IoT technology.

2. Objectives

- Design an IoT-based EV charging monitoring system.
 - Monitor charging voltage and temperature.
 - Simulate charging stages using a potentiometer.
 - Provide visual LED indications for different charging conditions.
 - Generate buzzer alerts during charging completion and fault conditions.
 - Display live charging data using the Blynk IoT platform.
 - Automatically stop charging when a fault is detected.
-

3. Hardware Components

- ESP32 Development Board
- DHT11 Temperature Sensor
- ZMPT101B Voltage Sensor
- 10k Potentiometer
- 4-Channel Relay Module
- Active Buzzer
- Green LED
- Yellow LED
- Two Red LEDs
- Breadboard
- Jumper Wires
- USB Cable



4. Software Used

- Arduino IDE
 - Blynk IoT Platform
 - GitHub
 - C++ Programming Language
-

5. System Workflow

1. ESP32 initializes all hardware modules.
 2. Temperature and voltage values are continuously monitored.
 3. Potentiometer simulates battery charging percentage.
 4. System status changes according to charging level:
 - System Ready
 - Charging
 - Charging Complete
 - Fault Protection
 5. LED indicators and buzzer provide visual and audible alerts.
 6. Live data is sent to the Blynk cloud dashboard.
 7. During a fault condition, the relay turns OFF to stop charging automatically.
-

6. LED Indication

System Ready

- Green LED ON

Charging

- Green LED ON
- Yellow LED ON

Charging Complete

- Green LED ON
- Yellow LED OFF
- Red LED ON
- Buzzer Beeps

Fault Condition

- Green LED OFF
 - Two Red LEDs ON
 - Buzzer ON
 - Relay OFF
-

7. Blynk Dashboard Features

- Voltage Monitoring
 - Current Display (Prototype Simulation)
 - Temperature Monitoring
 - Charging Power
 - Charging Status
 - Relay Status
 - System Ready Indicator
 - Charging Indicator
 - Charging Complete Indicator
 - Fault Indicator
-

8. Applications

- Smart EV Charging Stations
 - Residential EV Chargers
 - Educational IoT Projects
 - Industrial Charging Monitoring
 - Prototype Battery Management Systems
-

9. Advantages

- Real-time monitoring
 - Remote access through IoT
 - Automatic fault protection
 - Low-cost implementation
 - User-friendly dashboard
 - Easy to upgrade for commercial applications
-

10. Future Scope

- Real current measurement using SCT-013 sensor
 - Battery State of Charge estimation
 - AI-based fault prediction
 - Mobile notifications
 - Cloud database integration
 - Fast charging optimization
 - GPS-enabled charging station management
-

11. Conclusion

EV GUARD successfully demonstrates a smart IoT-based EV charging monitoring and protection system using ESP32 and Blynk IoT. The project provides continuous monitoring of charging conditions, remote dashboard visualization, automatic protection, and clear status indication through LEDs and buzzer alerts. The prototype serves as a scalable foundation for future smart EV charging infrastructure with enhanced monitoring, safety, and intelligent control.

Developed By

Haridharshan S

Bachelor of Engineering (Electrical and Electronics Engineering)

Project Name: **EV GUARD**